

Does AI-Generated Scientific Text Follow Human Argumentation Patterns? A CARS-Based Comparison of Research Article Introductions

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Abstract

Large language models are moving from helping write up research to helping do it, which makes it important to know how the scientific text they produce differs from human writing. Work on this question has stayed mostly at the surface, using lexical and stylistic cues that light paraphrasing erases. We look instead at rhetorical structure, the sequence of argumentative moves through which a text makes its case. We study research-article introductions under Swales' CARS model, and compare original introductions from published linguistics articles with generated counterparts of the same papers. We find that human-written introductions are more flexible in which moves they use and in what order, while the generated ones are more uniform. Giving the models the CARS definitions makes them more rigid.

1 Introduction

Large language models (LLMs) are increasingly involved throughout the scientific process, from proposing hypotheses and reasoning over the literature (Wang et al., 2023; Gottweis et al., 2026) to conducting entire discovery cycles on their own (Lu et al., 2024) and writing the papers through which the resulting research is communicated (Van Noorden and Perkel, 2023; Liang et al., 2024). As these models take on a greater role in scientific writing, it becomes important to understand how they build an argument and position the work relative to prior research, and whether they do so in the same way as human authors (Gao et al., 2023; Wu et al., 2025).

For comparing human and machine-generated text, prior work relies on lexical statistics, stylistic fingerprints, and syntactic patterns (Gehrmann et al., 2019; Ippolito et al., 2020; Mitchell et al., 2023). Such cues say little about how a model builds a text, and are less suitable for studying scientific writing where what carries the work is the

1 Introduction

Large language models ... research is communicated. As these models ... as human authors. For comparing human ... syntactic patterns. Such cues ... surface traces. We address this gap ... few paragraphs. We follow Swales' ... own introduction. We ask three research questions ... reliably (RQ1)? Expert CARS annotation ... to large corpora. Second, where do ... change (RQ3)? We answer these questions ... on the output. Our contributions are ... than human authors.

Figure 1: Swales' CARS steps on this paper's own introduction, cut into consecutive *segments*, each labeled with one CARS step. A segment's color *hue* represents the parent move, and the *shade* represents the step within the move. Move 1, establishing a territory (purple): **claiming centrality**, **topic generalizations**, **reviewing previous research**. Move 2, establishing a niche (magenta): **indicating a gap**. Move 3, occupying the niche (blue): **announcing present research**, **outlining purposes**, **announcing principal findings**.

argumentative structure. Moreover, models keep getting better at imitating human writing style, and light paraphrasing erases most surface traces (Krishna et al., 2023; Sadasivan et al., 2025).

We address this gap by looking below the surface, at the *rhetorical structure* of the text: the sequence of argumentative moves that fulfill coherent communicative functions within a text (Swales, 2004). We focus on research-article introductions (RAIs), because they are especially dense in such moves; an introduction has to argue the paper into existence in the space of a few paragraphs (Swales, 1990; Ozturk, 2007). We follow Swales' Create-A-Research-Space (CARS) model (Swales, 1990, 2004), a standard account of how introductions are organised. The CARS model describes an introduction through three rhetorical moves that typically progress from the general to the specific: the author establishes a research territory (Move 1), opens a

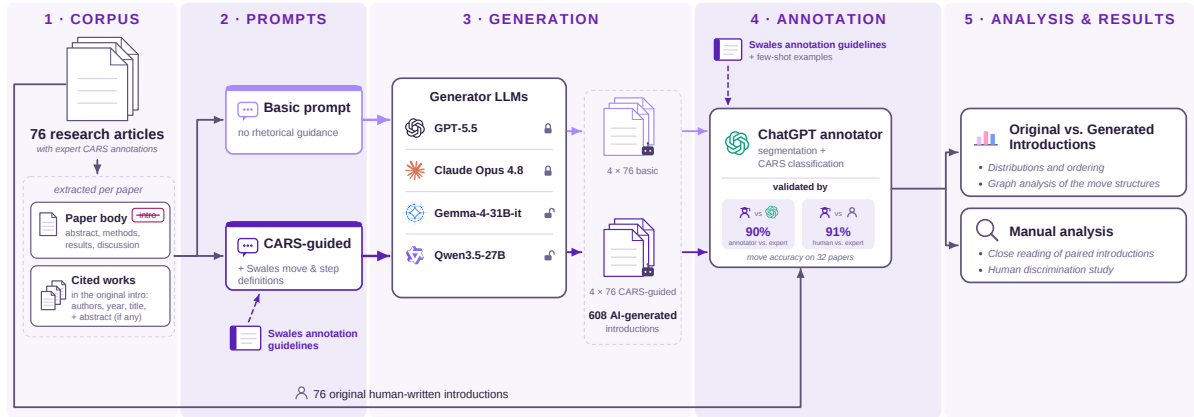


Figure 2: Overview of our framework. From each of the 76 corpus papers we reconstruct the content and the works cited in the original introduction. Four LLMs write each paper’s introduction under a basic and a CARS-guided prompt, giving 608 generated introductions. An automatic annotator then labels all original and generated introductions with CARS moves and steps. These labels feed the comparison of the original and generated introductions and the manual analysis.

niche in it, for example by pointing out a gap in previous work (Move 2), and occupies that niche with the present study (Move 3). Each move is realised through smaller rhetorical units called *steps*, which realise the move either individually or in combination (Biber et al., 2007), and in practice moves may recur, and appear in any order, and not every move need be present (D’Agostino et al., 2026); Figure 1 shows the model at work on this paper’s own introduction.

We ask three research questions. First, can an LLM-based annotator identify the CARS moves and steps of an introduction reliably (**RQ1**)? Expert CARS annotation is slow and hard to scale (Biber et al., 2007), so a dependable automatic annotator would open this kind of rhetorical analysis to large corpora. Second, where do human-written and LLM-generated introductions differ in their rhetorical structure (**RQ2**)? And what effect does explicit CARS guidance have on LLM-generated introductions: when an LLM is given the move and step definitions, does the rhetorical structure of the RAI it produces change (**RQ3**)?

We answer these questions in three parts (Figure 2). First, we build an automatic annotator that partitions an introduction into segments by identifying the boundaries of the rhetorical structures at the level of steps and labels each segment with its move and step (RQ1, §3.3); validated against two human annotators, ours reaches 92% of the human labelling accuracy and identifies moves and boundaries at close to human level. Second, we generate introductions at scale (§3.2). Starting from 76 pub-

lished linguistics articles whose introductions carry expert CARS annotations (D’Agostino et al., 2026). Four LLMs, two commercial and two open-weight, write every introduction twice, once from a basic prompt and once with the CARS move definitions added, giving 608 generated introductions. We use the annotator and human expertise to analyse how the generated introductions differ from the original ones (RQ2) and what influence CARS guidance has on the output (RQ3, §4).¹

Our contributions are:

- We propose to compare human-written and machine-generated research-article introductions by their rhetorical structure instead of surface-level features, and show that a human reader tells generated introductions from original ones in 90% of blind judgements by relying almost solely on these structures.
- We build and validate an LLM-based CARS annotator that segments and labels introductions, a reusable instrument for researchers who study rhetorical structure at scale.
- We present the first rhetorical structure comparison of original and LLM-generated RAIs, showing that generated introductions follow a narrower and more uniform range of move sequences than human authors.

¹We will release our code and data upon publication of this work.

2 Related Work

Two lines of work bear on our study: the generation and detection of machine-written scientific text, and the analysis of its rhetorical move structure.

2.1 LLMs for scientific writing and its detection

LLMs now draft many parts of a paper. They produce abstracts from a title (Gao et al., 2023), related-work and citation text (Li and Ouyang, 2024; Martin-Boyle et al., 2024), as well as whole papers, from science-specific models (Taylor et al., 2022) to autonomous pipelines that ideate, run experiments, and write (Lu et al., 2024). A growing fraction of recent papers contains LLM-modified text (Liang et al., 2024).

Detecting machine-generated text usually relies on surface signals: per-token probability rank (Gehrmann et al., 2019), probability curvature (Mitchell et al., 2023), decoding artifacts (Ippolito et al., 2020), and watermarking (Kirchenbauer et al., 2023). These signals are fragile. The text that best fools human readers is the hardest for detectors (Ippolito et al., 2020). For a strong enough model, the best possible detector approaches chance (Krishtna et al., 2023; Sadasivan et al., 2025). We therefore look past the surface to the rhetorical structure of the text, a deeper and possibly more robust signal that we show humans rely on when determining the source of the introduction.

2.2 Rhetorical move structure and its analysis

Genre analysis treats a RAI as a sequence of conventional rhetorical moves. Swales’ CARS model (Swales, 1990, 2004) is the well-established account in this regard: the author first establishes a territory, then carves out a niche, and finally occupies it. Recovering this move structure automatically has a long history. Argumentative Zoning assigns each sentence a rhetorical status (Teufel and Moens, 2002; Teufel et al., 2009; Teufel and Kan, 2011; Liakata et al., 2012). In parallel, work in English for Academic Purposes (EAP), the applied-linguistics field that teaches academic writing, has built classifiers for Swales’ moves and steps as feedback tools for novice writers (Anthony and Lashkia, 2003; Pendar and Cotos, 2008; Cotos et al., 2015; Dayrell et al., 2012), and more recent work pairs expert labels with transformer annotators (Li et al., 2024; Omotola et al., 2025). Closest to us, a recent strand compares AI-generated and human text by

move structure, for the abstract, where the main gaps are in move frequency and ordering (Kong and Liu, 2024; El-Dakhs et al., 2025). For the annotation itself, Kim and Lu (2024) prompts ChatGPT to label CARS moves and steps on applied-linguistics introductions.

We differ from these precedents in both terms of how we make the comparisons and how we annotate the texts. In our comparisons, we target the introduction rather than the abstract, as it is the rhetorical and argumentative gateway to a paper (Asadi and Rocci, 2026), compare several models rather than one, and generate each introduction from the cited literature, an operationalised approximation of the context a human author writes from. Studying rhetorical structure at scale also needs labels, and expert CARS annotation is slow and hard to scale, so a reliable automatic annotator is needed. We extend previous annotation methods Kim and Lu (2024) by benchmarking multiple providers, framing the task as joint segmentation and classification over the whole introduction instead of sentence-level labelling, and validating the annotator against a human–human ceiling.

3 Materials and methods

3.1 Corpus and gold annotations

Our starting point is the corpus collected by D’Agostino et al. (2026) and its manual expert annotations of rhetorical structure: 80 published research articles from Applied Linguistics journals, chosen as 20 papers per journal-prestige quartile (Q1–Q4). We first verified that every paper is publicly accessible online; four papers turned out to be available only behind paywalls, and were discarded. The final corpus comprises the remaining 76 papers.

We take these annotations, produced against a written annotation manual (Asadi, 2025), as our *gold* labels for validating the automatic annotator and answering RQ1. The scheme comprises three moves and ten steps (Table 4).

3.2 Generating introductions

Input context and preprocessing. A fair human-vs-AI comparison of *introductions* hinges on what the model is given. Prior genre comparisons prompt the model with only a short generic instruction (El-Dakhs et al., 2025) which is appropriate for an abstract, that summarises a paper the model can see, but not for an introduction. A human au-

thor writes an introduction from two sources: their own study, and the knowledge they take from the surrounding scholarly literature they have read and chosen to cite (Swales, 1990; Hyland, 1999; Samraj, 2005). We therefore give each model a reconstruction of the information context the author had: the paper’s title and content and the works cited in the original introduction, each with its bibliographic metadata and abstract where available. To assemble that context from the published PDFs, we apply a preprocessing pipeline (PDF-to-text conversion, manual correction, and the identification and enrichment of the works cited in each introduction), described in Appendix B. Each generated introduction is additionally targeted to the length of the paper’s original introduction, so that human–AI differences reflect how the model argues rather than how much it writes.

Models. We generate with four models that cover both commercial and open-weight systems: two widely used commercial models, GPT-5.5 and Claude Opus 4.8, and two open-weight models, Gemma-4-31B-it and Qwen3.5-27B, served locally. The open-weight condition matters because part of the research community runs models locally, for instance to avoid sharing unpublished ideas with commercial APIs. In the rest of the paper we refer to these models by short names: **ChatGPT**, **Claude**, **Gemma**, and **Qwen**. Appendix C gives the exact model versions and the settings of every model call.

Generation settings. To answer our RQ3, every model writes each introduction under two settings that differ *only* in rhetorical guidance. In the **basic** setting the model is asked to write the introduction directly, with no guidance about what an introduction should accomplish rhetorically. In the **CARS-guided** setting the prompt additionally supplies the Swales move and step definitions from the annotation manual and instructs the model to first plan the move sequence internally (moves may recur, interleave, and appear in any order) and then write the introduction realising that plan. Both system prompts are given in Appendix K.

Running the four models under both settings over the 76 papers yields $4 \times 2 \times 76 = 608$ RAIs.

3.3 CARS annotation

Automatic annotation. Comparing rhetorical structure between human-written and machine-generated introductions requires CARS annotations

on both sides of the comparison, and human annotation capacity does not scale to this setting: beyond the 76 human introductions, our study produces 608 generated ones (§3.2), each of which a trained annotator would have to segment and label. Building supervised classifiers would in turn require a large labelled training corpus (Li et al., 2024). Recent LLMs, however, perform annotation tasks competitively from instructions and a few demonstrations (Brown et al., 2020; Gilardi et al., 2023) and have shown promise specifically for rhetorical move–step analysis (Kim and Lu, 2024). We therefore build an LLM-based *automatic annotator* that recovers the gold annotation format by partitioning an introduction into segments and assigning each segment one of the ten step labels (§3.1).

Segmentation and classification. Recovering an annotation involves two decisions: *where* the rhetorical function changes (segmentation) and *which* function each segment realises (classification). Prompting the model to place boundaries alone systematically under-segments (mean 6.6 predicted vs. 9.6 gold segments on a pilot set²). Inspired by previous work on argumentative zoning (Teufel and Moens, 2002), we also tested classifying each sentence independently (with its immediate neighbors as context) and merging adjacent same-label sentences into segments; this removes boundary placement from the model’s job, but it over-segments (10.5 predicted segments/paper vs. 9.2 gold) because it is harder for the model to hold the same label across the several sentences that belong to one segment. Asking it to segment and label *jointly* improves boundary placement (pilot boundary F1 0.50 $\rightarrow$ 0.71). Our final formulation therefore hands the model the entire introduction and has it emit an ordered sequence of labeled segments with their classification.

Prompting strategy. The system prompt embeds the same annotation manual the human annotators followed (Asadi, 2025). The full prompt is given in Appendix K.4. We also included few-shot demonstrations (Brown et al., 2020) which are three fully worked gold annotations chosen to span the corpus’s granularity range: a coarse example (3 segments), a median one (9), and a fine-grained one (18). Within the prompt, we add a two-step decision procedure: identify the parent move first, then the step, and require a one-sentence rationale

²set of 16 papers that we did preliminary analysis on.

before each label, inspired by chain-of-thought prompting (Wei et al., 2022). Finally, we refine the prompt iteratively through error analyses on a held-out set of 10 introductions, which are used neither in the few-shot demonstrations nor in the annotator validation.

In preliminary experiments, we compared ChatGPT, Claude, and Gemini 3.1 Pro. ChatGPT was slightly ahead of Gemini 3.1 Pro and further ahead of Claude on step accuracy (0.773 vs. 0.762 and 0.708). We therefore use ChatGPT as the annotator for all analyses that follow.

Validating the annotator. Agreement with the expert gold annotations is hard to interpret on its own: CARS annotation is a judgement task on which trained humans themselves disagree (Teufel and Moens, 2002). We therefore measure the annotator *relative to a human baseline* obtained under identical conditions. Two graduate-student annotators with working knowledge of the CARS framework through the guidelines manual (Asadi, 2025), independently annotated 32 introductions from scratch (16 papers each, stratified 8 per journal quartile). Both the human annotations and the automatic annotator’s output are then scored against the expert gold (§3.1) on the same canonical texts.

We report two families of metrics one for segmentation and one for classification. Segmentation is scored with boundary F1, WindowDiff, and P_k ; classification is scored by the move and step F1. Appendix E.1 gives the full definitions.

Annotator performance The annotator reaches 92% of the human label accuracy (0.694 vs. 0.752; macro F1 0.664 vs. 0.699) and is essentially at the human baseline on structure: move accuracy 0.901 vs. 0.911 and boundary F1 0.725 vs. 0.756. Table 5 in Appendix D summarises the validation and gives the full per-class scores.

Given that the annotator identifies *moves* and places boundaries at close to human level, labels steps within 6 points of the human baseline, and errors mainly where humans also disagree, we adopt it as the single annotator for all introductions, human-written and generated, in the analyses that follow (§4. Using one annotator on both sides removes the annotator as a confound in the human–AI comparison.

4 Comparing Original vs. Generated Introductions

4.1 General distributions and ordering

We first ask whether the generated introductions reproduce the move and step *distributions* of the originals, and then whether they reproduce how the originals *order* and *select* their steps. Surface properties of the generated sets, such as their length, number of move segments, and mean segment length, are reported in Appendix G.

Move and step distributions. Table 1 reports, for each source, how its move segments distribute over the three moves and ten steps.

At the move level, the human authors devote 59.5% of their segments to establishing the territory (Move 1), 16.0% to establishing the niche (Move 2), and 24.5% to occupying the niche (Move 3). The generated introductions shift this balance from Move 1 toward Move 2. The Move-1 share in turn falls below the human level in five of the eight sets³ (down to 46.8%). This is the most consistent distributional difference we find.

At the step level, the human distribution is heavily skewed. A single step, *reviewing items of previous research*, accounts for 26.5% of all human segments, and together with claiming centrality (18.5%), topic generalization (14.5%), and outlining purpose (13.0%) they cover almost three quarters of the human segments. The generated sets keep the same rank order: reviewing previous research remains the largest step in every set, and outlining purpose stays within a few points of its human share (9.1–13.2% vs. 13.0%).

Finally, comparing each model’s basic and CARS-guided versions shows that the guidance works in the same direction for all four models, as it spends a smaller share of its segments on Move 1 (every Move-1 step shrinks for every model) and a larger share on Move 2 and Move 3. For three of the four models the guidance changes the move balance further away from the human one, adding to the over-use of Move 2 that the generated introductions already show.

Step ordering and selection Table 2 turns from what steps the introductions contain to how they are ordered. As full step sequences are diverse,

³A *set* is the 76 introductions written by one model under one prompt, so the four models and two prompts give eight generated sets; with the 76 originals this makes the nine sources compared throughout this section.

	<i>Original</i>	ChatGPT _b	ChatGPT _c	Claude _b	Claude _c	Gemma _b	Gemma _c	Qwen _b	Qwen _c
Move 1: Establishing a Territory	59.5	70.1*	62.9	57.5	51.1*	60.6	46.8*	54.4	48.3*
Claiming centrality (CC)	18.5	21.9*	20.4	19.2	17.3	19.9	17.1	16.0	15.8
Topic generalization (MTG)	14.5	22.3*	19.3*	14.0	11.5	15.7	9.9*	13.0	10.0*
Reviewing prev. research (RR)	26.5	25.9	23.1*	24.3	22.3*	25.0	19.9*	25.4	22.5*
Move 2: Establishing a Niche	16.0	14.6	17.9*	19.9*	22.8*	21.1*	22.1*	22.3*	26.6*
Continuing a tradition (CT)	3.6	3.8	4.3	3.8	4.1	6.6*	6.0*	4.6	4.3
Indicating a gap (Gap)	9.6	7.1	8.8	12.4*	13.7*	10.8	11.9*	13.0*	16.0*
Question raising (QR)	2.8	3.7	4.8*	3.7	4.9*	3.6	4.2	4.7*	6.3*
Move 3: Occupying the Niche	24.5	15.3*	19.2*	22.6	26.1	18.3*	31.1*	23.2	25.1
Announcing present res. (APR)	7.1	3.9*	5.2	7.4	6.7	3.3*	6.5	6.1	5.9
Announcing findings (APF)	3.2	2.3	3.9	3.1	5.2*	4.1	6.7*	6.0*	7.3*
Indicating structure (IS)	1.3	0.1	0.1*	0.5	1.1	0.0*	6.4*	0.7	0.3
Outlining purpose (OP)	13.0	9.1*	10.0*	11.6	13.2	10.9	11.5	10.4	11.6

Table 1: Distribution of the moves and steps: each cell is the percentage of all segments produced by the source (pooled over its 76 introductions) that carry the given label. *Original* is the human-written introduction. Subscripts: *b* = basic, *c* = CARS-guided prompting. * marks a value that differs significantly from the original’s (Appendix F).

Source	Top step 3-gram (%)	Cyclic %	Uniq. moves	Uniq. steps
<i>Original</i>	RR→MTG→RR (25)	66	2.67	5.3
ChatGPT _b	CC→MTG→RR (57)	70	2.93*	6.3*
ChatGPT _c	CC→MTG→RR (61)	63	3.00*	6.9*
Claude _b	RR→CC→RR (33)	74	2.95*	6.2*
Claude _c	CC→RR→Gap (42)	51	3.00*	6.6*
Gemma _b	RR→CC→RR (37)	67	2.96*	6.1*
Gemma _c	CC→RR→Gap (37)	50	3.00*	7.4*
Qwen _b	RR→Gap→RR (36)	95*	2.97*	6.8*
Qwen _c	CC→RR→Gap (34)	87*	2.99*	7.0*

Table 2: Step ordering profile. *Top step 3-gram*: the source’s most frequent sequence of three consecutive steps, followed in brackets by the percentage of the source’s introductions that contain it. *Cyclic %*: percentage of introductions that return to a *move* they had already left. *Uniq. moves* and *Uniq. steps*: mean number of *distinct* moves (out of 3) and of *distinct* steps (out of 10) that one introduction draws on.

we consider for each introduction its most frequent step 3-gram, whether it returns to a move it had already left (*Cyclic*), and how many distinct moves and steps it draws on. One habit is common to all nine sources: the most frequent opening step is everywhere the same, claiming centrality. A second commonality is what never occurs: the scheme’s eleventh step, Move-2 *counter-claiming*, is absent from the human gold annotations and appears in only 4 of the automatic annotator’s segments, so we exclude it throughout. This highlights that rather than attacking previous claims head-on, authors and models alike prefer the softer framing of indicating a gap.

Against the originals, the generated sets trade variety for uniformity. Humans open with the modal

centrality claim in only 46% of the papers, and another 32% open directly with the literature (RR), an opening the models almost never choose (0–12%); the models start with a centrality claim in 58–84% of their introductions. The humans’ most recurrent pattern, a Move-1-internal weave of reviewing and generalizing (RR→MTG→RR), appears in only a quarter of their papers, whereas each model has a stronger signature that appears at least 33% of the time.

Two thirds of the human introductions (66%) return at some point to a move they had already left, at a rate of 2.4 returns per introduction. Half of these returns (91 of 181) go back to Move 1, most often to review more literature after a first statement of the niche. The generated sets fall on both sides of that rate. Basic-prompted ChatGPT, Claude, and Gemma sit at the human rate (67–74%), and none of the three differs significantly from the originals of the same papers. Qwen departs from it upward, cycling in 95% (basic) and 87% (CARS-guided) of its introductions, significantly more than the humans. The CARS-guided sets depart downward, which we return to below (§5.1).

What further separates original from generated introductions is the selection: a human introduction uses on average 2.67 of the 3 moves and 5.3 of the 10 steps (28% of the originals omit an entire move), whereas most generated set realises all three moves and 6.1–7.4 steps. Where a human author selects from the scheme, the models cover it. Where the original establishes no niche (19 papers), states no explicit gap (32), or never claims centrality (10), 86–96% of its generated counterparts add the missing step.

Source	Similarity to the original				Within source	
	Edit _m	Edit _s	Bag _m	Bag _s	Self _s	Str _m
<i>Original</i>	–	–	–	–	0.33	36
ChatGPT _b	0.62	0.41	0.83	0.63	0.43	18
ChatGPT _c	0.61	0.41	0.81	0.62	0.44	9
Claude _b	0.61	0.42	0.81	0.63	0.40	24
Claude _c	0.61	0.43	0.80	0.61	0.46	11
Gemma _b	0.61	0.41	0.80	0.62	0.40	14
Gemma _c	0.60	0.39	0.76	0.57	0.46	8
Qwen _b	0.55	0.38	0.78	0.60	0.37	35
Qwen _c	0.56	0.38	0.75	0.58	0.44	25

Table 3: Graph comparison of move structures (Appendix E.2 defines the measures). Left: mean similarity of each generated RAIs to the same paper’s original, as edit similarity (Edit) and bag similarity (Bag), at the move (*m*) and step (*s*) level. Right: mean pairwise step edit similarity among the source’s own introductions (Self_s) and its number of distinct move structures (Str_m).

Comparing each model’s basic and CARS-guided variants shows the guidance pushing all four models in the same direction, toward more of the scheme being applied more uniformly, not toward the human profile. Guided introductions open with a centrality claim even more uniformly. The share of cyclical introductions falls for all four models (Claude 74 → 51%, Gemma 67 → 50%, ChatGPT 70 → 63%, Qwen 95 → 87%), and it falls past the human rate of 66%. Guidance also raises the number of distinct steps per introduction for every model.⁴

Finally, in Appendix H we use the journal-quartile data to show that the original introductions differ across venues while the generated sets do not reproduce that variation.

4.2 Graph analysis of the move structures

The comparisons above rest on pooled distributions; they say nothing about the similarity between individual introductions and disregard structure. Here we measure that structural similarity directly, between each generated set and the originals and within each set. Every introduction is treated as a labelled chain graph whose nodes are the CARS classes of the step segments in reading order, each node linked to the next. Two graphs are compared with two measures, each at the move (*m*) and step (*s*) level: *edit similarity* scores how far the two

class orders agree, and *bag similarity* scores how far the two introductions give their words to the same classes, with order excluded. Appendix E.2 defines both measures.

Which model’s introductions are most similar to the original ones? Table 3 (left) reports the mean similarity of each generated set to the same papers’ originals. Higher Bag than Edit similarities show that the models match what an original contains much better than how it orders it. What stands out is how low the edit similarities are in absolute terms: at the step level no generated set exceeds 0.43, so even the closest model reproduces well under half of the order in which the original author moves through the scheme.

How similar are a model’s introductions for different papers? Table 3 (right) reports each source’s self-similarity, the mean pairwise step edit similarity among its own 76 introductions, and the number of distinct move structures it realises. The originals are the most varied: their self-similarity (0.33) is below that of every generated set (0.37–0.46), and they realise 36 distinct move structures against the models’ 8–35. CARS guidance makes every model less diverse and more self-similar, following the findings of §4.1.

5 Manual analysis

5.1 Close reading of paired introductions

The statistics above say how often the generated structures deviate from the human ones, but not what a deviation looks like on the page. We therefore had five papers close-read by a senior graduate student with expertise in the CARS framework. Each paper was read as a triplet: the original introduction and Claude’s basic and CARS-guided versions of it, all shown with their automatic move annotations. The five span all four quartiles and differ in step edit similarity.

Findings The generated introductions are more rigidly ordered than the originals, in two senses.

The first is coverage. All ten generated introductions realize all three moves, while the originals do not: one of the five never marks a niche at all and passes from the territory straight to the present study. Where a human leaves the niche implicit or omits it, the models state it.

The second is order. The generated versions stay closer to the canonical Move 1–2–3 progression,

⁴This higher count is not greater diversity: each guided introduction covers more steps, but the guided sets repeat the same covering, which is why the number of distinct move structures they realise drops for every model (Table 3).

and where they leave it they leave it in one way, by returning from Move 2 to Move 1 to review more literature before restating the niche. Across the corpus, this single transition accounts for 89% of all returns in the generated sets but only 57% in the originals, which also go back to Move 1 after announcing the present study. The opening is the clearest case. Human authors sometimes open with the literature, a topic generalization, or even their purpose statement, and one of the five originals begins with Move 3. No generated introduction does, here or anywhere in the corpus: 0 of the 608 generated introductions open on Move 3.

One explanation is positional bias, that is, the models place the moves in the order in which the CARS model presents them rather than in the order the argument would require. A comparable pull toward the canonical order has been reported when LLMs are used to analyse moves rather than to write them (Yu et al., 2026). It is consistent with what CARS guidance does here: the prompt lists the moves in canonical order and says they may appear in any order, yet the introductions written under it follow the canonical order more closely than the basic ones (§4), the opposite of what the guidance was meant to achieve.

Authors who know the framework do not write more canonical introductions. A handful of the 76 articles themselves engage with Swales' CARS model or rhetorical move analysis, so their authors demonstrably know the framework. The same expert examined four of them and found that they follow no common structure. What separates them is the journal quartile (§H), not their knowledge of the framework. Appendix I discusses this in more detail.

5.2 What does a human reader rely on to distinguish between original and generated introductions?

We ask whether the structural differences reported above can be put to use, that is, whether a reader who is given the CARS model guidelines and a summary of the corpus-level insights can tell an original introduction from a generated one. The study is blind and was carried out by a student researcher from outside the project, who judged 120 introductions one at a time, 60 originals and 60 Claude generations of the same 60 papers. Appendix J gives the full protocol.

The judgements rest on structure, not on style.

The reader classified 108 of the 120 (90%) introductions correctly, and named the argumentation structure as a basis for the decision on 118 of the 120 items against the writing style on 15 (more than one basis could be selected per item). They also report having segmented each introduction into moves in their head, and relying on the relations between the moves on top of the moves themselves, that is, the order and the returns that §4.1 shows the two sources—human and models—handle differently. They also weighed the corpus-level tendencies they had been given beforehand about which moves the models over-use and which the human authors favour. The accuracy therefore comes from the rhetorical structure rather than from the writing, and the differences highlighted in §4 can be used to distinguish between original and generated RAIs.

Introductions written with the CARS-guided prompt are the easier ones to spot.

Half of the 60 generated introductions were produced with the CARS-guided prompt and half with the basic one. The reader identified all 30 CARS-guided ones as machine-written, against 24 for the basic. This supports, from the reader's side, what §4.1 found in the structures themselves: the CARS-guided prompt makes the introductions more uniform and more rigid in their ordering, and that uniformity is what makes them recognisable as machine-written.

6 Conclusion

We compared LLM-written RAIs with the human-written originals of the same papers by their rhetorical structure, using a validated automatic annotator of Swales' CARS moves. The generated introductions are more rigid, and rigid in the same ways across all four models used for generation. Where a human author selects some moves, the models cover most of it, and they settle into a narrower range of move sequences. Supplying the CARS definitions pushes the models further from human structure. A reader with knowledge of CARS framework and a summary of models' patterns could discriminate between original and generated RAIs with 90% accuracy, with higher performance for RAIs generated using CARS-guided. Rhetorical structure therefore carries a usable trace of machine authorship, one that sits in the argument rather than in the surface style that paraphrasing erases.

Limitations

Fidelity of the input context. The information context we give the models (§3.2) is an operational approximation of what the author had. First, the cited works are represented by their bibliographic metadata and abstracts, while the author read the full works. Second, the citation list is taken from the finished introduction, so the models inherit the author’s selection of the literature; a human author performs that selection as part of writing. These choices make the comparison feasible and consistent across papers, but the models write from a curated subset of the author’s knowledge, not from the same position.

Single discipline and corpus scale. Our study covers a single domain, applied linguistics, and a corpus of 76 papers released by D’Agostino et al. (2026). The scarcity of hand-labelled data is in fact what motivated the automatic annotator we propose (§3.3): rather than annotate every introduction by hand, we validate an LLM annotator against human labels and use it to scale the analysis to material that would otherwise be out of reach. All text is also recovered from PDFs by automatic extraction, which introduces noise; like Teufel and Kan (2011) we operate on extracted scholarly text, and we mitigate the noise by manual quality control of the parsed introductions. A larger, multi-domain corpus would strengthen the statistical claims and let us test how far the patterns hold beyond this discipline.

Generalisability of the CARS model. The CARS model is not a fixed, universal template: the move and step inventory and the frequency of individual steps vary across disciplines and even sub-disciplines, which bounds how far our findings transfer beyond applied linguistics. This is visible in our own gold annotations, where the counter-claiming step is effectively unused. Variability persists within the discipline as well: Ozturk (2007) finds that two applied-linguistics sub-disciplines employ “different and almost unrelated” move structures. Because our corpus pools sub-disciplines, it may contain heterogeneous move structures, and the results should not be read as a single canonical “applied linguistics” introduction structure. The more transferable finding is therefore the *gap* between human and AI move use, not the absolute distributions; the same comparison in another field could yield different human baselines.

Depth of the argumentation analysis. Swales’ CARS model captures the rhetorical moves an introduction makes, but it is not a deep model of argumentation: it labels the function of a passage (establishing a territory, a niche, or occupying it) without analysing the internal structure of the reasoning, such as the claims, the evidence, and the warrants that link them. We treat this work as a first step. Our results support that reading: even at the level of moves, the framework already surfaces systematic differences between human and AI introductions, which shows that Swales’ scheme gives usable hints and directions about where the two diverge. Finer-grained argumentation frameworks, for example the Argumentum Model of Topics (Rigotti and Greco, 2019) or a pragma-dialectical analysis (van Eemeren, 2018), could be layered on top of this analysis in future work to move from where an argument is made to how it is constructed, and so yield more detailed findings about the quality of machine-generated scientific writing.

Annotator and ceiling. We use an LLM as the move annotator and validate it against two human annotators who labelled 32 introductions between them, 16 papers each (§3.3). The human–human ceiling is thus estimated from a subset of the corpus, and no introduction was labelled by both annotators. A larger double-annotated set would sharpen the ceiling.

Ethics Statement

Source material. Every paper in our corpus is a published research article that is publicly accessible online. We checked this for each paper before including it, and we dropped the four papers whose full text was only available behind a paywall (§3.1). No unpublished, or otherwise private material was collected or processed. We use the published text only for analysis, and we do not redistribute the articles themselves.

Annotation work and compensation. The human annotations used in this paper (§3.1, §3.3) come from three kinds of contributors. Some of the annotators are authors of this paper, and their annotation work is part of their own research. Others are colleagues who took part voluntarily as academic collaborators and received no separate payment for this task. One annotator is a research assistant in our lab, employed on a salary that covers their working time, of which this annotation was a part.

All annotators knew what the annotations would be used for and took part with that understanding.

Data handling in the tools we used. Two of the four generation models, Gemma and Qwen, were served locally, so their inputs never left our own machines. The two commercial models were accessed through the OpenAI and Anthropic APIs, whose terms state that data sent through the API is not used to train their models. The same holds for the model we use as an automatic move annotator. Because all of our inputs are published papers, nothing we sent to an external service was confidential in the first place.

Generated introductions. We generate introductions to mimic a use case that is already common in research writing, so that we can study how the resulting argumentation differs from human writing. This is a research artifact and not a writing service: none of the generated text is intended for submission anywhere. We also note that several journals and conferences restrict or forbid automatically generated content in submitted work.

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A The CARS annotation scheme

Table 4 lists the three CARS moves and ten steps of the annotation scheme (§3.1).

Move	Steps
M1: Establishing a Territory	Claiming centrality (CC); Topic generalization (MTG); Reviewing previous research (RR)
M2: Establishing a Niche	Continuing a tradition (CT); Indicating a gap (Gap); Question raising (QR)
M3: Occupying the Niche	Announcing present research (APR); Announcing principal findings (APF); Indicating structure (IS); Outlining purpose (OP)

Table 4: The Swales CARS move/step taxonomy used for annotation (3 moves, 10 steps). The acronym after each step is the short form used in the other tables, written as move-step (e.g. M1-CC).

B Preprocessing pipeline

This appendix details the preprocessing pipeline (§3.2) that assembles each paper’s generation context from the published PDFs.

PDF-to-text conversion. We converted every paper’s PDF to structured text with Docling (Auer et al., 2024),⁵ selected over pdfminer.six⁶ and PyMuPDF⁷ after manually inspecting the candidate extractions: Docling best preserved the multi-column layouts, section headings, and tables. Structured reference lists, for which layout-oriented extractors perform poorly, were parsed separately with GROBID (Lopez, 2009).⁸

Manual correction. We then iterated over the extracted texts and manually fixed the remaining minor extraction mistakes (broken hyphenation, mis-ordered text blocks, and running headers or footers leaking into the body), yielding a clean full-text version of all 76 papers.

Introduction citations and their abstracts. We next identified every work cited in each *introduction*: in-text citations were matched to reference-list entries automatically using author-year citation patterns, and the result was then manually verified and corrected paper by paper. Across the corpus this yields roughly 1,065 cited works (a mean of

⁵<https://github.com/docling-project/docling>

⁶<https://github.com/pdfminer/pdfminer.six>

⁷<https://github.com/pymupdf/PyMuPDF>

⁸<https://github.com/kermitt2/grobid>

Annotator	Classification			Segmentation
	step-acc	step-F1	move-acc	bound-F1
HB	0.752	0.699	0.911	0.756
AA	0.694	0.664	0.901	0.725

Table 5: The automatic annotator (AA) vs. the human baseline (HB), both scored against the expert gold on the same 32 introductions. The two column groups score the two decisions separately. *Classification* scores which rhetorical function a piece of text realises: step-acc and step-F1 are the accuracy and the macro-averaged F1 of the per-sentence step label, and move-acc is the same accuracy after aggregating each sentence’s step to its parent move. *Segmentation* scores where the rhetorical function changes, independently of the labels: bound-F1 is boundary F1 with a ± 5 -character tolerance. Appendix E.1 gives the full definitions.

14 per introduction). Finally, we retrieved bibliographic metadata and abstracts for these works from multiple scholarly databases: OpenAlex,⁹ Crossref,¹⁰ and Semantic Scholar,¹¹ complemented by a cascade of further academic APIs for hard-to-find entries. Abstracts were recovered for about 72% of the cited works; for the remainder, we enter the generation context with their title and authors only.

C Model configuration

The introductions (§3.2) were generated with gpt-5.5, called through the OpenAI Batch API on 7 July 2026, claude-opus-4-8, called through the Anthropic Message Batches API on 6 July 2026, and google/gemma-4-31B-it and Qwen/Qwen3.5-27B, served locally with vLLM (Kwon et al., 2023).¹² The automatic annotator (§3.3) is gpt-5.5, also through the Batch API, and the preliminary annotator comparison additionally used gemini-3.1-pro-preview. Every call uses the provider’s default decoding settings and low reasoning effort.

D Annotator validation results

Table 5 reports the validation of the automatic annotator against the human baseline (§3.3).

Table 6 gives per-class F1 for the ten CARS steps under the per-sentence label metric, for the human second annotators and the automatic anno-

⁹<https://openalex.org>

¹⁰<https://www.crossref.org>

¹¹<https://www.semanticscholar.org>

¹²<https://github.com/vllm-project/vllm>

Class	Human	Annotator
M1-CC	0.70	0.46
M1-MTG	0.63	0.55
M1-RR	0.86	0.84
M2-CT	0.23	0.32
M2-Gap	0.77	0.77
M2-QR	0.42	0.35
M3-APR	0.62	0.63
M3-APF	0.77	0.62
M3-IS	0.92	0.92
M3-OP	0.85	0.81

Table 6: Per-class F1 on the 32-paper validation set, scored against the expert gold. *Human* is the second-annotator baseline; *Annotator* is our automatic annotator. Classes are named move-step as in Table 4.

tator, both scored against the expert gold on the 32 validation introductions (§3.3). The human–annotator gap is concentrated in claiming centrality (M1-CC) and, to a lesser extent, making topic generalization (M1-MTG) and announcing principal findings (M3-APF); the two sides are at parity on the two most frequent classes (M1-RR, M3-OP) and on indicating a gap (M2-Gap).

E Evaluation metrics

E.1 Annotator validation metrics

These are the metrics used to validate the automatic annotator against the human baseline (§3.3). They form two families that decouple *where* the moves change from *which* move each segment realises.

Segmentation (label-independent). **Boundary F1** matches predicted to gold boundaries within a ± 5 -character tolerance. We also report **WindowDiff** (Pevzner and Hearst, 2002) and P_k (Beeferman et al., 1999), the standard text-segmentation error scores that credit near-miss boundaries gracefully.

Classification. We compute classification metrics at the sentence level rather than the segment level, because the predicted and gold segments can differ in number; this is also standard practice in the literature on argumentative zoning and CARS move/step classification (Teufel and Moens, 2002; Teufel and Kan, 2011; Pendar and Cotos, 2008; Cotos et al., 2015; Li et al., 2024). Each sentence in the introduction is assigned the predicted and gold labels of the segment it belongs to, and we report sentence-level label accuracy. Finally, we aggregate each sentence’s step to its parent move and report *move accuracy*.

E.2 Move-structure similarity measures

We define the two measures used in the graph analysis (§4.2). Both compare two introductions through the chain graphs of their move segments, and both are computed at the move level (3 classes) and the step level (10 classes).

Edit similarity. Edit similarity scores how far two introductions order their classes the same way:

$$\text{EditSim}(a, b) = 1 - \frac{\text{Lev}(a, b)}{\max(|a|, |b|)},$$

with Lev the Levenshtein distance (Levenshtein, 1966) between the class sequences a and b : the minimum number of insertions, deletions, and substitutions that turns one sequence into the other, normalised by the longer sequence’s length so the score lies in $[0, 1]$. Because the graphs are chains, the Levenshtein distance between the label sequences is exactly the graph edit distance between the graphs (Sanfeliu and Fu, 1983). Before the comparison, consecutive segments carrying the same class are merged into one node (the analysis convention of §G).

Bag similarity. Bag similarity ignores the order and asks only how much of an introduction goes to each class. Every introduction divides its words among the classes, which gives one share per class; the shares add up to one and form a distribution p . Two introductions are then compared by how much of these shares they spend the same way:

$$\text{BagSim}(p, q) = \sum_c \min(p_c, q_c),$$

that is, for each class we keep the smaller of the two shares and add these up, which is the part of the two profiles that overlaps. If one introduction gives 60% of its words to Move 1 and the other gives 40%, that class contributes 0.4 to the sum. The score is 1 when the two introductions divide their words over the classes in exactly the same way and 0 when they have no class in common. This is the histogram intersection of Swain and Ballard (1991), known in statistics as the overlapping coefficient (Weitzman, 1970; Inman and Bradley, 1989); for two distributions it equals one minus the total variation distance between p and q .

A class’s share is its share of the introduction’s *words*, not its share of the segments. The reason is that the same text can be cut into more or fewer segments: an annotator that splits one

long stretch of literature review into three segments would make that class look three times as important if we counted segments, while its word share does not move. Read together, the two measures separate content from arrangement: a pair with high bag similarity but low edit similarity gives its words to the same rhetorical classes at similar weights but orders them differently.

F Significance testing

All comparisons between an original and a generated set are paired by paper: the two introductions compared are always of the same paper, so the paper is the unit of analysis ($n = 76$). We test them with the Wilcoxon signed-rank test (Wilcoxon, 1945), except for cyclicity, which is a per-introduction binary and is tested with the McNemar test (McNemar, 1947). A star marks $q < 0.05$.

The model comparison of §4.2 is tested the same way, on each paper’s step edit similarity to its own original.

G Descriptive statistics

Table 7 compares the original human-written introductions with the eight generated sets on three surface properties: length, number of segments, and mean segment length, with every set annotated by the automatic annotator (§3.3).

For the generation length, ChatGPT overshoots the target by roughly 20% while the other models stay close. The clearest structural difference is in rhetorical granularity. Throughout the analyses, consecutive segments that carry the same step label are counted as one segment, so a new segment always marks a change of rhetorical function. Every generated set realises more segments than the human originals (9.9–13.9 segments per introduction vs. the human 9.4), meaning the models switch rhetorical function more often than human authors. All sets except CARS-guided Claude also use shorter segments (50.2–57.3 words per segment vs. the human 65.1), while CARS-guided Claude develops its segments at human length (66.1). CARS guidance moves Claude and Qwen visibly closer to the human granularity (e.g. Claude 11.4 $\rightarrow$ 9.9 segments), but leaves ChatGPT and Gemma essentially unchanged.

Source	Words /intro	Segments /intro	Words /segment
<i>Original (human)</i>	<i>615</i>	<i>9.4</i>	<i>65.1</i>
ChatGPT, basic	743	13.9	53.5
ChatGPT, CARS	745	13.7	54.4
Claude, basic	655	11.4	57.3
Claude, CARS	653	9.9	66.1
Gemma, basic	563	10.5	53.7
Gemma, CARS	590	10.5	56.0
Qwen, basic	659	13.1	50.2
Qwen, CARS	612	11.7	52.4

Table 7: Descriptive statistics of the original and generated introductions (76 each): mean words per introduction, mean number of segments, and mean words per segment.

H Variation across journal quartiles

The corpus is stratified by journal quartile (Q1–Q4, §3.1), which lets us ask whether the rhetorical profile of the originals varies with publication venue, and whether the generated introductions follow that variation. Table 8 reports, per quartile, how much of the CARS repertoire an introduction uses (the mean number of distinct moves and distinct steps) and how often it cycles (the percentage of introductions that return to an already-visited move). The original introductions differ across quartiles: Q1 papers use almost all three moves (2.88 on average) and cycle the most (75%), while Q3 papers use the fewest moves (2.50) and cycle the least (55%); Q2 papers draw on the widest range of steps (5.7 of the 10).

The generated introductions show much less variation. In every quartile they use nearly all three moves (2.97–2.99), more steps than the originals (6.1–7.2 vs. the human 5.0–5.7), and cycle at a nearly constant rate (66–74%, against the human range of 55–75%). The quartile differences visible in the originals are therefore not reproduced. This is expected: the models write in a uniform style, and we gave them no information about the journal quartile of the paper they were writing for.

I The Swales-awareness effect

A handful of the 76 articles themselves engage with Swales’ CARS model or rhetorical move analysis, so their authors demonstrably know the framework. Does that knowledge show in how they build their own introductions? The expert of §5.1 examined four such articles, one from each journal quartile. This appendix gives the readings summarised there.

Knowing the framework does not make these au-

	Source	Uniq. moves	Uniq. steps	Cyclic %
Q1	Original	2.88	5.0	75
	Generated	2.99	6.1	73
Q2	Original	2.65	5.7	70
	Generated	2.97	7.2	74
Q3	Original	2.50	5.3	55
	Generated	2.97	6.7	66
Q4	Original	2.70	5.3	65
	Generated	2.98	6.5	66

Table 8: CARS repertoire and cyclicity by journal quartile. *Uniq. moves*: mean number of *distinct* moves (of 3) per introduction; *Uniq. steps*: mean number of *distinct* steps (of 10) per introduction; *Cyclic %*: percentage of introductions that return to a *move* they had already left. *Generated* pools the eight generated sets.

thors write more canonical introductions: none of the four follows the standard Move 1–2–3 funnel. Two never mark an explicit niche and pass from the territory straight to their purpose statement. A third does raise a niche but folds it into a territory statement instead of giving it its own move, so a niche that is rhetorically present is not counted as Move 2. A fourth returns to the territory after announcing its work and ends on Move 1. Their command of the model is real, but it does not surface as a more schematic structure.

What does track structure is the journal quartile, an association also reported across high- and low-impact venues (Alanazi and Alqarni, 2022). In our corpus (§H), higher-quartile introductions move among the three moves more freely and realize Move 3 more fully, whereas lower-quartile ones lean on Move 1 and compress or omit the later moves. The quartile is a stronger predictor of an introduction’s structure than the author’s knowledge of the CARS model.

J Human detection study

This appendix details the detection study of §5.2, in which an informed but blind reader judges, one introduction at a time, whether it was written by a human or generated by a model.

Reader. The judge is a student researcher who is not involved in the project and knows nothing about its design, corpus, or hypotheses. Before starting they read only the Swales CARS annotation guidelines and the insight summary described below. They are blind throughout to item provenance, to the human-to-AI ratio, and to the pairing

of the two versions of each paper.

Sampling and design. Sixty papers were sampled from the 76 in the corpus, stratified by journal quartile (Q1 12, Q2–Q4 16 each). Each sampled paper contributes two items, its original introduction and one generated introduction for the same paper, giving 120 items in total. The generated introductions all come from Claude, which is among the models whose structures come closest to the human originals (§4.2). Two factors are crossed at the paper level, so both of a paper’s items share the same cell: the prompting setting of the generated item (basic vs. CARS-guided) and whether the pair is shown with automatic move annotations (labelled vs. unlabelled). Each of the four cells holds 15 papers, which yields 60 items shown with move annotations (30 original + 15 basic + 15 CARS-guided) and 60 shown as plain text. The 120 items are presented in a single fixed order in which a paper’s two versions are always at least 20 positions apart, so that the two versions of one paper are never close enough for the pairing to become visible.

Insight summary shown to the judge. Before starting, the judge reads the annotation guidelines and a short summary of the corpus-level tendencies from §4.1 and §4.2.

Response. Figure 3 shows the interface. For each introduction the judge answers a provenance question on a five-point scale (1 sure it is human, 2 most likely human, 3 cannot decide, 4 most likely AI, 5 sure it is AI) and a required question on the basis of the decision (the argumentation or CARS structure, the writing style, or a guess; multiple may be selected). In the labelled condition the automatic move annotations appear as coloured highlights over the text, and the key above the introduction lists only the moves that occur in the item on screen.

Analysis. The five-point ratings are read as a binary judgement by splitting the scale at its midpoint, 1 and 2 counting as a vote for human and 4 and 5 as a vote for AI. We report the number of correct judgements and the accuracy. The reader never used the middle point of the scale.

Results. Table 9 reports the study. The reader was correct on 108 of the 120 items, and the two kinds of error are balanced: six generated introductions passed as human and six originals were taken

Rhetorical moves in this text (automatic CARS annotation):

Move 1 — Establishing a territory: M1-CC Claiming centrality M1-RR Reviewing previous research

Move 2 — Establishing a niche: M2-Gap Indicating a gap

Move 3 — Occupying the niche: M3-OP Outlining purpose(s)

M1-RR Contemporary instructed language learning approaches emphasize on encouraging learners to actively take part in communicative tasks (Ellis, 2014). However, despite efforts made by teachers to provide such opportunities, it is very common for them to face learners who are reluctant to engage in verbal communication; which is due to various reasons (Zulkepli, Othman, & Khaja, 2013). The issue of reticence occurs not only among less proficient learners, but also the proficient ones such as preservice ESL teachers (Fahim & Dhamotharan, 2016; Yousef, Jamil, & Razak, 2013; Zulkepli, Tajuddin, AlBakri, Atan, & Bakar, 2019). Empirical studies have long proven the strong link between speaking and learning a target language (Goh, 2014; Long, 2015; Swain, 2000). M1-CC This has triggered interest among scholars in the area of willingness to communicate (WTC); resulting in numerous empirical studies that set to investigate the variables that affect this psychological construct (Pawlak, Mystkowska-Wiertelak, & Bielak, 2016; Shirvan, Khajavy, MacIntyre, & Taherian, 2019; Zhang, Beckman, & Beckman, 2018). M2-Gap However, studies done on this area predominantly focus on in-class WTC where learners normally engage in interaction with "familiar interlocutors, such as teachers and peers, in relatively predictable L2 scenarios" (Lee & Hsieh, 2019, p. 64) To date, there is scanty research on out-of-class WTC where learners will speak with "unfamiliar acquaintances or strangers in somewhat unpredictable situations" (Lee & Hsieh, 2019: 64); which will be more challenging even for motivated and advanced learners. M3-OP Thus, the purpose of the study is to identify the link between motivation and WTC among a group of advanced learners i.e., pre-service ESL teachers.

Q1 — Who wrote this introduction?

☐ 1 — Sure it's human^[1]
☐ 2 — Most likely human^[2]
☐ 3 — Can't decide^[3]
☒ 4 — Most likely AI^[4]
☐ 5 — Sure it's AI^[5]

Q2 — What did you base your decision on? (tick all that apply)

☐ Argumentation / Swales CARS structure^[6]
☐ Writing style^[7]
☐ Just a guess^[8]

Optional comment

anything that struck you...

#34

Submit

Figure 3: The annotation interface used in the human detection study (§5.2). A single introduction is shown, with its automatic CARS move annotations as coloured highlights in the labelled condition, above the five-point provenance question and the required decision-basis question. The colour key at the top lists only the moves that occur in the item on screen. The judge sees the whole introduction by scrolling; the text is cut off at the foot of the screen here.

Condition	Items	Correct
All items	120	108 (90%)
<i>By move annotation (all items)</i>		
Labelled	60	54 (90%)
Unlabelled	60	54 (90%)
<i>Generated items, by prompting setting</i>		
Basic	30	24 (80%)
CARS-guided	30	30 (100%)

Table 9: Results of the human detection study (§5.2). *Labelled/Unlabelled*: whether the item was shown with its automatic CARS move annotations as coloured highlights. *Basic/CARS-guided*: the prompting setting of the generated item. *Correct* counts judgements that match the true provenance, so the percentage is the accuracy.

for machine writing.

Prompting setting. All 30 CARS-guided generations were identified as machine-written against 24 of the 30 basic ones. The uniformity behind the effect is visible in these 120 items as well as in the corpus at large: 63% of the CARS-guided generations follow the plain $1 \rightarrow 2 \rightarrow 3$ move order against 13% of the basic ones and 15% of the originals, and the 30 CARS-guided introductions realise only 6 distinct move structures where the 60 originals realise 29.

Showing the move annotation made no difference. The reader was correct on 54 of 60 items in the labelled condition and on 54 of 60 in the unlabelled one.

We read this as a statement about what a reader needs rather than as evidence that the annotations carry nothing. The reader named the structure as the basis of the judgement on 58 of the 60 unlabelled items, and reports having done the move analysis in their head, from the content alone, on the items that came without annotations. The analysis was therefore being done either way. What the reader had been given beforehand was the annotation scheme and a short summary of how the two sources differ across the corpus, and that turned out to be enough to act on.

K Prompts

This appendix reproduces all prompts used in the study: the system prompts of the two generation settings (§3.2), the per-paper user message they share, and the system prompt of the automatic annotator (§3.3).

K.1 Basic generation prompt

The two generation settings share the same role framing, input description, output rules, and JSON output shape. The basic setting uses the following system prompt verbatim.

Basic Generation Prompt (system)

You are an expert academic writer specialising in applied linguistics and English language teaching (EFL/ESL). Your task is to write the **Introduction** section of a research article.

You are given:

1. The article's content – its abstract and body sections (literature review, methodology, results, discussion). This tells you what the study is about, what it did, and what it found. **IMPORTANT:** the original Introduction has been withheld on purpose; do not try to recall or reconstruct it – write a fresh one from the content you are given.
2. The works the original authors cited in their introduction. Each entry gives the authors, year and title, and includes an abstract where one was available (some entries have only the title and authors – judge from the title what the work is about).

Rules:

- Do NOT output a section heading – only the introduction prose.
- Cite sources in author-year format, e.g. “Smith & Jones, 2019”. Use the works listed above where relevant; you may also cite other works where the writing calls for it.
- Length: write to the target length stated with the article below and stay within the given range. Do not pad or over-elaborate to reach it, and do not substantially exceed it.

Return ONLY a JSON object with this exact shape (no markdown, no commentary):

```
{"introduction": "<the full introduction text>"}
```

K.2 CARS-guided generation prompt

The CARS-guided setting extends the basic system prompt with the move-planning instruction below and the CARS move and step definitions, taken from the annotation manual (Asadi, 2025) and inserted where [CARS MOVE AND STEP DEFINITIONS] appears.

CARS-Guided Generation Prompt (system, additions)

You will write the introduction following Swales' CARS (Create A Research Space) model, whose moves and steps are defined below.

Before writing, PLAN THE MOVES INTERNALLY: think through which CARS moves and steps the introduction will use and in what sequence. The moves do NOT have to appear in 1 → 2 →

3 order – they recur and interleave freely, and any ordering is acceptable (a move can be revisited, e.g. returning to a Move 1 review after a Move 2 gap, and a real introduction may open or close on any move). Keep this plan entirely to yourself.

Then WRITE the introduction that realises your internal plan. Output ONLY the introduction prose – never the plan, and no move/step labels or headings in the text.

[CARS MOVE AND STEP DEFINITIONS] (the move and step definitions of the annotation manual, Asadi, 2025)

K.3 Generation user message

In both generation settings, each paper is supplied in a user message containing the article title, the article content, the cited works, and the target length.

Per-Paper Generation User Message

Article title

{title}

Article content (abstract + body; the original introduction is withheld)

{body}

Key prior works cited in the original introduction

{references}

Target length

{length_instruction}

Now produce the Introduction. Return only the JSON object described in your instructions.

K.4 Annotator prompt

The system prompt of the automatic annotator (§3.3) consists of the task framing (segmentation rules, the two-step labelling procedure, the output format, and the numbered label set), the full annotation manual (Asadi, 2025), the operational clarifications distilled from the error analyses of §3.3, and the few-shot demonstrations. The box below shows the structure of the prompt, with each detailed part summarised by a bracketed placeholder. Each introduction is then supplied in a short user message (“Please segment the following research article introduction into coherent argumentative units. Return only the JSON — no explanation.”).

Annotator Prompt (system)

You are an expert in academic discourse analysis, specialising in Swales' CARS (Create a Research Space) model for research article introductions.

Task. Following the annotation manual below: (1) split the introduction into coherent argumentative segments; (2) for each segment,

first identify its MOVE, then its step; (3) give a one-sentence rationale and output the class number.

[SEGMENTATION INSTRUCTIONS] (five rules: cover the whole introduction, never cut mid-sentence, segments verbatim and in order, and a new segment whenever the communicative function changes)

[TWO-STEP LABELLING PROCEDURE] (decide the move first — Move 1/2/3 with their class-number ranges — then the step within that move)

[OUTPUT FORMAT] (a JSON “segments” array whose elements carry the verbatim “text”, a one-sentence “rationale” written before deciding the label, and an integer “label”)

[LABEL SET] (the numbered list of the eleven steps, grouped by move)

[ANNOTATION GUIDELINES] (the full annotation manual of [Asadi, 2025](#), inserted verbatim)

[OPERATIONAL CLARIFICATIONS] (tie-breaker rules distilled from the error analyses of §3.3: the decision order within Move 1, granularity and recurrence, one communicative function per segment, continuing a tradition vs. indicating a gap, frameworks and approaches, and purpose vs. present research)

[WORKED EXAMPLES] (three gold-annotated demonstration introductions — coarse, median, and fine-grained — rendered in the output format, plus a constructed contrast paragraph for the three Move-1 steps)

Guiding principle. Always classify segments by the communicative function of each sentence – what the author is doing rhetorically at that point in the introduction – not by surface cues, sentence boundaries, the presence of connectors, or paragraph layout. A new segment is warranted only when the communicative function genuinely changes.